From Estimation to Hypothesis Testing

Until now, our primary goal has been **estimation**: using sample data to guess the value of an unknown parameter θ, either with a single point or an interval.

Often, however, the objective of an investigation is not to estimate a parameter, but to decide whether a specific claim about the parameter is true.

Examples:

- Is the mean waiting time of a queue strictly less than 50 seconds?
- Is a new manufacturing process producing a lower variance of defects than the old process?

To make these decisions using random sample data, we use the formal framework of **Statistical Hypothesis Testing**.

The Null and Alternative Hypotheses

A statistical hypothesis is a statement about the probability distribution of a population. We always formulate two competing hypotheses:

1. **The Null Hypothesis (H_0):**
   - Represents the status quo, historical precedent, or "no effect."
   - It is the hypothesis assumed to be true unless the data provides overwhelming evidence against it.
   - Mathematically, it always contains the equality condition ($=, \geq, \leq$).
2. **The Alternative Hypothesis (H_1 or H_A):**
   - Represents the claim we are actually trying to establish.
   - Mathematically, it requires a strict inequality ($\neq, >, <$).

Simple and Composite Hypotheses

These terms describe how many possible population distributions a hypothesis allows:

Simple Hypothesis

Specifies exactly one probability distribution (all relevant parameters are fixed). For example, if $X_i \sim N(\mu, \sigma^2)$ with known σ^2, then $H_0 : \mu = \mu_0$ is simple.

Composite Hypothesis

Allows more than one possible probability distribution. In the same model, $H_1 : \mu > \mu_0$ is composite because it allows many values of μ. Likewise, $H_0 : \mu \leq \mu_0$ is composite.

An equality restriction is simple only when it fully specifies the distribution, including any nuisance parameters.

Decision Errors

Because we are making decisions based on a random sample, we will occasionally make mistakes. There are two specific types of errors:

- **Type I Error:** Rejecting the null hypothesis when it is actually true. (A "false positive").
- **Type II Error:** Failing to reject the null hypothesis when it is actually false. (A "false negative"). The probability of avoiding a Type II error, $1 - \beta$, is the power of the test.

The Significance Level (α):
We construct tests to strictly control the probability of a Type I error at a predetermined low level, denoted by α (commonly 0.05 or 0.01):

$$\alpha = \mathbb{P}(\text{Type I Error}) = \mathbb{P}(\text{Reject } H_0 \mid H_0 \text{ is true})$$

Power and the Outcomes of a Test

A test's decision and the true state of nature combine to produce four possible outcomes:

Decision	**State:** H_0 true	**State:** H_1 true
Reject H_0	Type I error α	Correct rejection Power $= 1 - \beta$
Fail to reject H_0	Correct decision $1 - \alpha$	Type II error β

Error Probabilities and Power

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}), \qquad \beta = P(\text{fail to reject } H_0 \mid H_1 \text{ true})$$

$$\text{Power} = P(\text{reject } H_0 \mid H_1 \text{ true}) = 1 - \beta$$

For composite alternatives, power depends on the true parameter value.

Test Statistic and Rejection Region

To execute the test, we compute a **test statistic** from our sample data. The exact probability distribution of this statistic *under the assumption that H_0 is true* must be known.

The **Rejection Region (Critical Region)** is the set of values for the test statistic for which we will reject H_0.

The Logic:

- If the calculated test statistic falls into the rejection region, the observed data is highly unlikely to have occurred if H_0 were true. We reject H_0 in favor of H_1.
- If it does not, we do not have enough evidence to reject H_0. *(Note: We say "fail to reject H_0", we never say "accept H_0").*

Example: Testing a Normal Mean (Known Variance)

Suppose $X_1, \ldots, X_n \sim \text{Normal}(\mu, \sigma^2)$ with known σ^2. We want to test if the mean is strictly greater than a specific baseline value μ_0.

Hypotheses:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu = \mu_1 > \mu_0$$

Test Statistic: Under the assumption that $\mu = \mu_0$ (the boundary of H_0), we know:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \text{Normal}(0, 1)$$

Rejection Region: For a significance level α, we reject H_0 if the sample mean is unusually large. Specifically, we reject H_0 if our computed test statistic Z satisfies:

$$Z \geq z_\alpha$$

Power when $\mu = \mu_1$: Under this alternative, $Z \sim N\left(\frac{(\mu_1 - \mu_0)\sqrt{n}}{\sigma}, 1\right)$, so

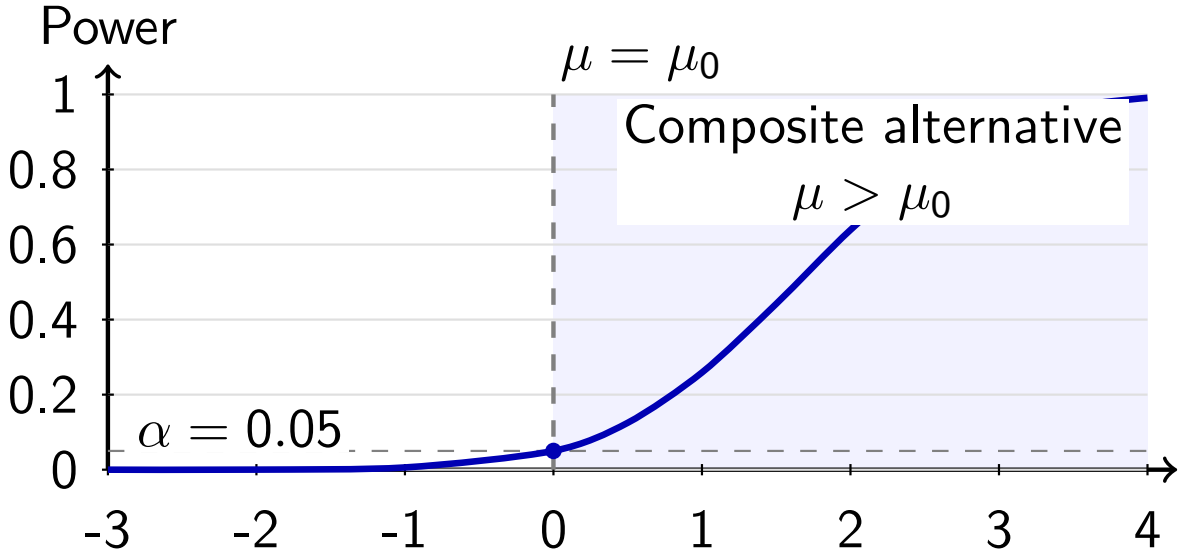
$$\text{Power} = P_{\mu_1}(Z \geq z_\alpha) = 1 - \Phi\left(z_\alpha - \frac{(\mu_1 - \mu_0)\sqrt{n}}{\sigma}\right).$$

For $\alpha = 0.05$, $z_\alpha \approx 1.645$.

Power Curve for a Composite Alternative

For the right-tailed known-variance Z-test at $\alpha = 0.05$, power varies with the true mean μ:

$$\pi(\mu) = 1 - \Phi(z_\alpha - \delta), \qquad \delta = \frac{(\mu - \mu_0)\sqrt{n}}{\sigma}.$$



Standardized effect $\delta = (\mu - \mu_0)\sqrt{n}/\sigma$

At the null boundary $\mu = \mu_0$, power equals α; it increases as μ moves farther into the alternative.

Biased and Unbiased Tests

For a level-α test, compare its power under alternatives with its size:

Definitions

- **Unbiased:** Power is at least α for every value in the alternative.
- **Biased:** Power falls below α for at least one alternative.

The Trivial Randomized Test

A data-independent random decision rejects H_0 with probability α (equivalently, accepts it with probability $1 - \alpha$), regardless of the sample. Its power is exactly α for every alternative. An unbiased test has power at least α for every alternative, so it is never worse than this baseline.

When the Alternative Is Very Close

The Right-Tailed Normal Z-Test

Its power function is

$$\pi(\mu) = 1 - \Phi\left(z_\alpha - \frac{(\mu - \mu_0)\sqrt{n}}{\sigma}\right).$$

At $\mu = \mu_0$, $\pi(\mu_0) = \alpha$. For every $\mu > \mu_0$, power exceeds α; therefore, this test is unbiased for the right-sided alternative.

For fixed n and σ, the standardized separation is $\delta = (\mu_1 - \mu_0)\sqrt{n}/\sigma$. As μ_1 approaches μ_0, the two sampling distributions become harder to distinguish:

$$\lim_{\mu_1 \downarrow \mu_0} \text{Power}(\mu_1) = \alpha.$$

No fixed-size test can keep high power against alternatives arbitrarily close to the null. Increasing n makes a fixed mean difference easier to detect.

Example: Testing a Normal Mean (Unknown Variance)

In practice, the true population variance σ^2 is rarely known. We must estimate it using the sample variance S^2, which introduces extra uncertainty into our test.

Hypotheses:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu = \mu_1 > \mu_0$$

Test Statistic: Replacing σ with S changes the sampling distribution. From our previous derivations, we know this ratio forms a Student's t-distribution. Under the boundary assumption $\mu = \mu_0$:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

Rejection Region: For a significance level α, we reject H_0 if our computed test statistic falls into the extreme right tail of the t-distribution:

$$T \geq t_{\alpha,n-1}$$

Note: $t_{\alpha,n-1}$ is the critical value such that $P(T > t_{\alpha,n-1}) = \alpha$.

Directionality: One-Sided vs. Two-Sided Tests

Two-Sided (Two-Tailed) Test

Used when we care about deviations in **either direction** from the baseline.

- **Hypotheses:** $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$
- **Rejection Region:** The error rate α is split in half. We reject if the test statistic is unusually large or unusually small: $|Z| \geq z_{\alpha/2}$

One-Sided (One-Tailed) Test

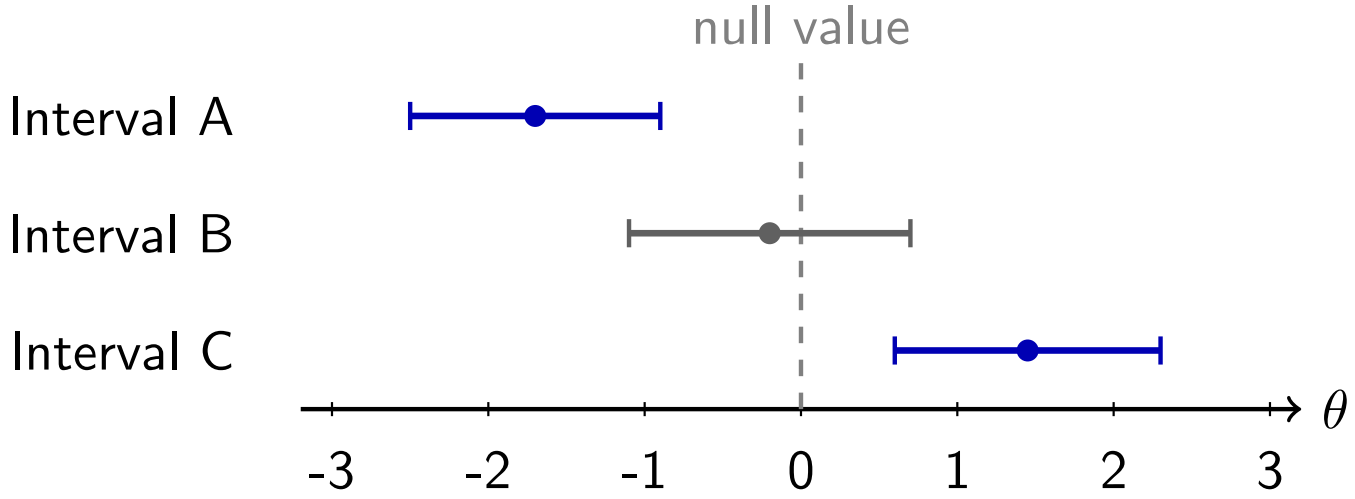
Used when we strictly care about a deviation in **one specific direction** (e.g., verifying if a new process is strictly *faster*, not just different).

- **Right-Tailed:** $H_1 : \mu > \mu_0 \implies$ Reject H_0 if $Z \geq z_\alpha$
- **Left-Tailed:** $H_1 : \mu < \mu_0 \implies$ Reject H_0 if $Z \leq -z_\alpha$

Trade-off: A one-sided test makes it easier to reject H_0 in the chosen direction ($z_\alpha < z_{\alpha/2}$), but makes it impossible to detect a change in the opposite direction.

Confidence Intervals and Hypothesis Tests

For a two-sided test of $H_0 : \theta = 0$ at $\alpha = 0.05$, compare the 95% confidence interval with the null value:



- Intervals A and C exclude 0: reject H_0 at the 5% level.
- Interval B includes 0: fail to reject H_0.

For a two-sided test, this correspondence holds when the confidence interval and test use the same model and assumptions.

Notation for a Statistical Testing Problem

Let $X = (X_1, \ldots, X_n)$ be the random sample, x its observed value, and $\theta \in \Theta$ the unknown parameter.

Hypotheses

$$H_0 : \theta \in \Theta_0 \qquad \text{vs.} \qquad H_1 : \theta \in \Theta_1 = \Theta \setminus \Theta_0.$$

A test is a rule $\phi(x) \in [0, 1]$, the probability of rejecting H_0 after observing x; for a nonrandomized test, $\phi(x)$ is 0 or 1.

Size and Power

$$\text{Size} = \sup_{\theta \in \Theta_0} E_\theta[\phi(X)] \leq \alpha, \qquad \pi_\phi(\theta) = E_\theta[\phi(X)].$$

For $\theta \in \Theta_1$, the Type II error probability is $\beta(\theta) = 1 - \pi_\phi(\theta)$.

Earlier Tests in the General Notation

Each rejection rule is a test function $\phi(X)$: it equals 1 when we reject H_0 and 0 otherwise.

Test	**Null set Θ_0**	**Rejection rule $\phi(X)$**
Known-variance, right-tailed Z	$\{\mu : \mu \leq \mu_0\}$	$\mathbf{1}\{Z(X) \geq z_\alpha\}$
Unknown-variance, right-tailed t	$\{(\mu, \sigma^2) : \mu \leq \mu_0,\ \sigma^2 > 0\}$	$\mathbf{1}\{T(X) \geq t_{\alpha,n-1}\}$

In the one-sided examples, $\mu = \mu_0$ is the boundary of the composite null $\mu \leq \mu_0$; the critical value is calibrated there.

For the Z-test: $$\pi_\phi(\mu) = E_\mu[\phi(X)] = 1 - \Phi\left(z_\alpha - \frac{(\mu - \mu_0)\sqrt{n}}{\sigma}\right).$$

Constructing a Test: The Likelihood Approach

How do we systematically build a test statistic for complex scenarios where standardization isn't obvious? The most powerful and universal method relies on the **Likelihood Function**.

The Setup

Let $X_1, \ldots, X_n$ be a sample with joint probability density $f(x \mid \theta)$. The likelihood function is $L(\theta \mid x) = f(x \mid \theta)$.

The Intuition:

1. Compare how well the data is explained when θ is restricted to Θ_0 versus when it can range over the full parameter space Θ.
2. If the data at hand is more likely when the θ values lie in Θ_0 then we favor H_0, else we dont.

The Likelihood Ratio Test (LRT)

Compare the best likelihood under the null restriction with the best likelihood over the full parameter space.

Likelihood Ratio Statistic

$$\lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta \mid x)}{\sup_{\theta \in \Theta} L(\theta \mid x)}.$$

Interpretation

The numerator is no larger than the denominator, so $0 \leq \lambda(x) \leq 1$. Small values indicate that the data fit much better somewhere outside Θ_0 than within it.

Decision rule: Reject H_0 when $\lambda(x) \leq c$, with c chosen to control the Type I error probability at α.

A Simple-vs-Simple Normal Mean LRT

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, with known σ^2. Test two fixed values:

$$H_0 : \mu = \mu_0 \qquad \text{vs.} \qquad H_1 : \mu = \mu_1, \quad \mu_1 > \mu_0.$$

Both hypotheses are **simple**. For this simple-vs-simple test, the likelihood ratio is

$$\Lambda(\mathbf{x}) = \frac{L(\mu_0 \mid \mathbf{x})}{L(\mu_1 \mid \mathbf{x})} = \exp\left[-\frac{n(\mu_1 - \mu_0)}{\sigma^2}\left(\bar{x} - \frac{\mu_0 + \mu_1}{2}\right)\right].$$

Since $\mu_1 - \mu_0 > 0$, this ratio decreases as $\bar{x}$ increases. The LRT therefore rejects H_0 for sufficiently large values of $\bar{X}$.
Hence,

$$\phi(x) = \mathbf{1}(\bar{X} > c)$$

We also want, $\mathbb{E}_{H_0}(\phi(\bar{X})) = \alpha$ to fix the size of the test to α., So we need to find c such that $P(\bar{X} > c) = \alpha$.

The Simple LRT Gives the One-Sided Z-Test

Choose the cutoff to make the test have significance level α. Under H_0,

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1), \qquad P_{\mu_0}(Z \geq z_\alpha) = \alpha.$$

Thus the level-α rejection region is

$$\bar{X} \geq \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} \qquad \Longleftrightarrow \qquad Z \geq z_\alpha.$$

Equivalently, choose $c = \Lambda\big(\bar{x} = \mu_0 + z_\alpha \sigma/\sqrt{n}\big)$; then $\Lambda(\mathbf{X}) \leq c$ gives exactly this rejection region.

So for any fixed $\mu_1 > \mu_0$, the simple-hypothesis LRT yields the same one-sided Z-test cutoff. It is most powerful for that specified simple alternative.

Neyman–Pearson Lemma

Statement

For simple $H_0 : f = f_0$ versus simple $H_1 : f = f_1$, the size-α likelihood-ratio test that rejects for small $f_0(\mathbf{X})/f_1(\mathbf{X})$ is most powerful among all tests with size at most α. Randomization at the cutoff can be used if needed to attain exact size.

- The preceding test is most powerful for its specified simple alternative $\mu = \mu_1$.
- For a composite null, control size over the whole null: $\sup_{\theta \in \Theta_0} E_\theta[\phi(X)] \leq \alpha$.
- For a composite alternative, the lemma alone does not give one test that is most powerful at every alternative. Additional structure, such as a monotone likelihood ratio, can establish stronger results.

The Composite One-Sided Normal-Mean LRT

Now test the composite hypotheses in the same known-variance model:

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad H_0 : \mu \le \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0.$$

The generalized likelihood ratio compares the best fit under the null with the unrestricted best fit:

$$\lambda(\mathbf{x}) = \frac{\sup_{\mu \le \mu_0} L(\mu \mid \mathbf{x})}{\sup_{\mu \in \mathbb{R}} L(\mu \mid \mathbf{x})}.$$

The unrestricted MLE is $\hat{\mu} = \bar{x}$; the null-constrained MLE is $\tilde{\mu}_0 = \min(\bar{x}, \mu_0)$. Hence

$$\lambda(\mathbf{x}) = \begin{cases} 1, & \bar{x} \le \mu_0, \\ \exp\left[-\dfrac{n(\bar{x} - \mu_0)^2}{2\sigma^2}\right], & \bar{x} > \mu_0. \end{cases}$$

The Composite LRT Gives the Same Z-Test

Write $Z = (\bar{X} - \mu_0)/(\sigma/\sqrt{n})$. Then

$$\lambda(\mathbf{X}) = \exp\left[-\frac{\{\max(Z, 0)\}^2}{2}\right].$$

Viewed as a function of the observed z, $\lambda(z) = 1$ for $z \leq 0$, while for $z > 0$, $\lambda'(z) = -z\lambda(z) < 0$. Thus, λ is nonincreasing overall and strictly decreasing in the upper tail.
For $0 < \alpha < 1/2$, set $z_\alpha = \Phi^{-1}(1 - \alpha)$ and choose $c = \exp(-z_\alpha^2/2)$. The LRT rule becomes

$$\lambda(\mathbf{X}) \leq c \quad \Longleftrightarrow \quad Z \geq z_\alpha.$$

This is exactly the one-sided Z-test. Over the composite null, the rejection probability is largest at $\mu = \mu_0$ and equals α there; it is no larger for $\mu < \mu_0$.

Constructing the LRT for Component Lifetimes

Let's construct the formal Likelihood Ratio Test (LRT) for the lifespan of n independent components, $X_i \sim \text{Exponential}(\theta)$.

Hypotheses: $H_0 : \theta = \theta_0$ (baseline) vs. $H_1 : \theta \neq \theta_0$

The Likelihood Function

The PDF is $f(x \mid \theta) = \frac{1}{\theta} e^{-x/\theta}$. For n independent observations, the joint likelihood is the product:

$$L(\theta \mid \mathbf{x}) = \prod_{i=1}^{n} \frac{1}{\theta} e^{-x_i/\theta} = \theta^{-n} e^{-\frac{1}{\theta}\sum x_i} = \theta^{-n} e^{-Y/\theta}$$

where $Y = \sum_{i=1}^{n} X_i$ is our total accumulated lifetime.

Evaluating the Likelihood Ratio $\lambda(Y)$

1. The Denominator (Unrestricted Maximum)

To maximize $L(\theta)$ over all possible values, we find the Maximum Likelihood Estimator (MLE) via standard calculus. For the Exponential distribution, the MLE is the sample mean: $\hat{\theta} = \frac{Y}{n}$.

$$\sup_{\Theta} L(\theta) = \left(\frac{Y}{n}\right)^{-n} e^{-\frac{Y}{Y/n}} = \left(\frac{Y}{n}\right)^{-n} e^{-n}$$

2. The Numerator (Restricted Maximum)

Under H_0, there are no parameters to estimate; θ is locked to θ_0.

$$\sup_{\Theta_0} L(\theta) = \theta_0^{-n} e^{-Y/\theta_0}$$

The Ratio:

$$\lambda(Y) = \frac{\theta_0^{-n} e^{-Y/\theta_0}}{(Y/n)^{-n} e^{-n}} = \left(\frac{Y/n}{\theta_0}\right)^{n} e^{n-\frac{Y}{\theta_0}}$$

Simplifying the Rejection Region

The LRT tells us to reject H_0 when $\lambda(Y) \leq c$. But how do we define that in terms of our actual data? Let's look at the shape of the ratio.

Recall our sample mean is $\hat{\theta} = Y/n$. We can rewrite our ratio in terms of $\hat{\theta}$:

$$\lambda(\hat{\theta}) = \left(\frac{\hat{\theta}}{\theta_0}\right)^n e^{n\left(1-\frac{\hat{\theta}}{\theta_0}\right)}$$

If you plot this function against $\hat{\theta}$, it has a very specific geometry:

- It reaches its absolute maximum (where $\lambda = 1$) exactly when $\hat{\theta} = \theta_0$.
- As $\hat{\theta}$ moves away from θ_0 in **either direction**, the function strictly and rapidly **decreases** towards 0.

The Intuitive Conclusion

Small λ means $\hat{\theta}$ is far from θ_0, giving a **two-tailed test**. Since $\lambda(Y)$ rises to $Y = n\theta_0$ and then falls, it is nonmonotone in Y. Thus this LRT is not uniformly most powerful: the Neyman–Pearson lemma covers simple alternatives, while MLR yields optimality for one-sided alternatives.

LRT for Discrete Events: Poisson Distribution

Let's apply the LRT to a discrete scenario. Suppose we observe the number of network server errors over n distinct hours: $X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$.

Simple Hypotheses: We test a baseline rate against a known elevated rate (where $\lambda_1 > \lambda_0$).

$$H_0 : \lambda = \lambda_0 \quad \text{vs.} \quad H_1 : \lambda = \lambda_1$$

The Likelihood Ratio

The joint likelihood is $L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}\lambda^{Y}}{\prod x_i!}$, where $Y = \sum X_i$.
Because both hypotheses are simple, the ratio is just $L(\lambda_0)$ over $L(\lambda_1)$:

$$\Lambda(Y) = \frac{e^{-n\lambda_0}\lambda_0^{Y}}{e^{-n\lambda_1}\lambda_1^{Y}} = e^{-n(\lambda_0 - \lambda_1)} \left(\frac{\lambda_0}{\lambda_1}\right)^{Y}$$

Simplifying the Poisson Rejection Region

The LRT rule states we reject H_0 if $\Lambda(Y) \leq c$. Let's analyze the geometry of our ratio to find a practical cutoff.

$$\Lambda(Y) = e^{-n(\lambda_0 - \lambda_1)} \left(\frac{\lambda_0}{\lambda_1} \right)^Y$$

The Monotonicity Argument

- We assumed the alternative rate is higher: $\lambda_1 > \lambda_0$.
- Therefore, the fraction $\left(\frac{\lambda_0}{\lambda_1} \right)$ is strictly less than 1.
- A fraction less than 1 raised to the power of Y strictly **decreases** as Y gets larger.

The Practical Test

Because $\Lambda(Y)$ strictly decreases as Y increases, saying *"reject when $\Lambda(Y)$ is unusually small"* is identical to saying *"reject when Y is unusually large"*.

$$\text{Reject } H_0 \text{ if } Y \geq k$$

The Discreteness Penalty

We want to find an integer cutoff k to strictly bound our Type I error rate to $\alpha = 0.05$. Under H_0, our sum is $Y \sim \text{Poisson}(n\lambda_0)$.

The Problem with Discrete Distributions

Because Y only takes exact integer values, the cumulative tail probabilities "jump" in discrete steps.

Imagine looking at a Poisson table and finding:

- $P(Y \geq 15 \mid H_0) = 0.072$ (Violates $\alpha \leq 0.05$)
- $P(Y \geq 16 \mid H_0) = 0.031$ (Safe, but leaves 0.019 unused!)

The Consequence: We are forced to choose $k = 16$. This makes our test **conservative** (actual $\alpha = 0.031$). By shrinking the rejection region, we make it harder to reject H_0, reducing the **power** of our test.

A Binomial LRT: Testing a Success Probability

Suppose n independent trials each succeed with probability p, and let $K = \sum_{i=1}^{n} X_i$ be the number of successes, where $X_i \sim \text{Bernoulli}(p)$. Test a baseline probability against a specified larger one:

$$H_0 : p = p_0 \qquad \text{vs.} \qquad H_1 : p = p_1, \quad p_1 > p_0.$$

Since $K \sim \text{Binomial}(n, p)$, the likelihood ratio is

$$\Lambda(K) = \frac{L(p_0)}{L(p_1)} = \left(\frac{1-p_0}{1-p_1}\right)^n \left(\frac{p_0(1-p_1)}{p_1(1-p_0)}\right)^K.$$

The factor raised to K is less than 1, so $\Lambda(K)$ decreases as the number of successes increases. Thus the LRT rejects H_0 for sufficiently large K. Choose the integer cutoff using $K \sim \text{Binomial}(n, p_0)$ under H_0. if the discrete tail skips α, the actual level of the test will be smaller, which in turn will hurt the power of the test.

Recovering Power: The Randomized Test

To use the remaining α while maximizing power, define a **Randomized Test Function**, $\phi(y)$, as the probability of rejecting H_0 after observing y.

Constructing the Exact Test

Instead of a hard threshold, we define our rejection rule as:

- If $Y > 15$, we **reject** H_0 ($\phi = 1$), as in the cutoff $Y \geq 16$.
- If $Y < 15$, we **fail to reject** H_0 ($\phi = 0$).
- If $Y = 15$, we reject H_0 with probability γ.

We choose γ to perfectly hit our exact $\alpha = 0.05$ budget:

$$\alpha = P(Y > 15) + \gamma \cdot P(Y = 15)$$

$$\gamma = \frac{\alpha - P(Y > 15)}{P(Y = 15)} = \frac{0.05 - 0.031}{0.041} \approx 0.463$$

Relative to the cutoff $Y \geq 16$, power increases by $\gamma P_{\lambda_1}(Y = 15) > 0$. By Neyman–Pearson, this exact-size test is most powerful for the specified $\lambda_1 > \lambda_0$.

Beyond Binary Decisions: The Need for Nuance

So far, we have used a fixed significance level α (e.g., 0.05) to create a strict boundary. If our statistic crosses the line, we reject H_0.

The Limitation of the Fixed Cutoff

Suppose our $\alpha = 0.05$ rejection region is $Z \geq 1.645$:

- A test statistic of $Z = 1.65$ triggers a rejection.
- A test statistic of $Z = 5.50$ also triggers a rejection.

Both yield the exact same binary conclusion ("Reject H_0"), but $Z = 5.50$ provides vastly stronger mathematical evidence against the null.

How do we report the *strength* of our evidence so that other scientists can judge it for themselves?

Defining the p-value

Instead of asking "Did we cross the α line?", we invert the perspective and ask: *"If the null hypothesis were perfectly true, how likely is it to see data this extreme?"*

Definition

The p-value The p-value is the probability of obtaining a test statistic **at least as extreme** as the one that was actually observed, assuming that the null hypothesis (H_0) is true.

Mathematical Formulation (Right-Tailed Test)

If T is our random test statistic, the rejection region is $\{T > c\}$ and t_{obs} is the specific number we just calculated from our sample data:

$$p\text{-value} = P(T \geq t_{obs} \mid H_0 \text{ is true})$$

Using the p-value in Practice

The p-value acts as the lowest possible α level at which we could reject H_0. This gives us a universal decision rule for any software output.

The Universal Decision Rule

Compare the computed p-value directly against your chosen α:

- If $\mathbf{p} \leq \alpha$: The data is sufficiently extreme. **Reject** H_0.
- If $\mathbf{p} > \alpha$: The data is not extreme enough. **Fail to reject** H_0.

Crucial Misinterpretations to Avoid

- The p-value is **NOT** the probability that H_0 is true. It evaluates $P(T > t_{obs} \mid H_0)$, completely different from $P(H_0 \mid T = t_{obs})$.
- A large p-value does **NOT** prove that H_0 is true; it only means we currently lack the evidence to dispute it.

Comparing Two Populations: Known Variances

Often, we want to compare two independent groups. Let $X_1, \ldots, X_{n_1} \sim N(\mu_1, \sigma_1^2)$ and $Y_1, \ldots, Y_{n_2} \sim N(\mu_2, \sigma_2^2)$.

Hypotheses: We test the difference between the true means, $\Delta = \mu_1 - \mu_2$.

$$H_0 : \mu_1 - \mu_2 = \delta_0 \quad \text{vs.} \quad H_1 : \mu_1 - \mu_2 \neq \delta_0$$

(Most commonly, $\delta_0 = 0$ to test for strictly identical means).

The Two-Sample Z-Test

By the linearity of expectation and variance for independent variables, the difference in sample means is normally distributed:

$$E[\bar{X} - \bar{Y}] = \mu_1 - \mu_2 \quad \text{and} \quad Var(\bar{X} - \bar{Y}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

Standardizing this gives us our test statistic:

$$Z = \frac{(\bar{X} - \bar{Y}) - \delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

Unknown Variances: The Assumption of Equality

Assume equal but unknown population variances: $\sigma_1^2 = \sigma_2^2 = \sigma^2$.

The Pooled Variance Estimator

Pool the two sample variances, weighted by their degrees of freedom:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

The Pooled Two-Sample t-Test

The pooled two-sample statistic is:

$$T = \frac{(\bar{X} - \bar{Y}) - \delta_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

Notice the total degrees of freedom: $v = n_1 + n_2 - 2$.

Beyond Means: Testing Categorical Distributions

So far, we have tested specific parameters (like μ or λ). But what if we want to test if our data follows an **entire distribution**?

Suppose we have categorical data grouped into k distinct bins (e.g., days of the week, blood types, or histogram bins for a continuous variable).

- **Null Hypothesis (H_0):** The data comes from a specific theoretical distribution (the probabilities for each of the k bins are exactly $p_1, p_2, \ldots, p_k$).
- **Alternative (H_1):** The data does not follow this distribution.

Observed vs. Expected

For a sample size of n, we count the **Observed frequency** in each bin (O_i).

If H_0 is true, the **Expected frequency** in each bin should be:

$$E_i = n \cdot p_i$$

Pearson's Chi-Square Statistic

We need a test statistic that aggregates the total error between our reality (O_i) and our theory (E_i).

The Test Statistic

Karl Pearson introduced the following statistic to measure this discrepancy:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

Why divide by E_i?

We square the error $(O_i - E_i)^2$ so negative and positive errors don't cancel out. But an error of 10 is massive if you only expected 5 events, and completely negligible if you expected 10,000 events!

Dividing by E_i scales the error relative to the bin size.
If that total number of samples follows a Poisson distribution then bin counts follow Poisson distribution, the variance equals the mean E_i. We are standardizing by the variance!

The Distribution and Rejection Region

As the sample size $n \to \infty$, Pearson proved that this statistic converges perfectly to the distribution we already know:

$$\chi^2 \sim \chi^2_{k-1}$$

We have k bins, so why do we lose a degree of freedom? Because the total sample size n is fixed ($\sum O_i = n$). If you know the counts of the first $k-1$ bins, the final bin's count is locked. You only have $k-1$ independent pieces of information.

The Rejection Region

Because the test statistic squares all the errors, a perfectly fitting model yields $\chi^2 = 0$. Bad fits yield large positive numbers.

Therefore, this is **always a right-tailed test**! We reject H_0 if:

$$\chi^2 \geq \chi^2_{\alpha,k-1}$$